

Thesis Synopsis
On
Presence and Abundance of Parasites in Two Different Cultured
Freshwater Fish Species

Submitted By

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ID:18207030

Session:2021-2022

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Introduction:

Aquaculture has emerged as a vital sector for meeting the increasing demand for fish protein globally. In Bangladesh, freshwater aquaculture contributes significantly to the nation's economy and food security, with major cultured species. However, the sustainability and productivity of aquaculture systems are often hampered by the prevalence of parasitic infections, which can adversely affect fish health, growth rates, and market value (Rahman et al., 2017). Parasites in aquaculture systems not only result in direct losses due to fish mortality but also lead to sub-lethal effects such as reduced feed conversion efficiency and increased susceptibility to secondary infections (Woo, 2006).

Parasites are important components of host biology, population structure and indeed ecosystem functioning. They can be found in any fish species and within any type of aquatic and culture system (Marcogliese, 2004). They range from protozoans to metazoans including myxozoans, trematodes, cestodes, acanthocephalans, nematodes, and crustaceans (Marcogliese, 2004). The knowledge of the status of parasite diversity in the tropics is still inadequate (Dobson et al., 2008).

The Food and Agricultural Organization of the United Nations (2009) reported that to satisfy an increasing demand in freshwater fish, extensive research must include studies of their parasites for optimal production levels. As, The major diseases associated with fish are parasites, bacteria, viruses and toxic algae that reduces fish production by affecting the normal physiology of fish and if left uncontrolled, it can result in mass mortalities or in some cases, can be serve as source of infection for human and other vertebrates that consumed fish (Paperna, 1996). It is usually the parasitic infections that predispose fish to secondary infections like fungal, bacterial and viral disease (Sarig, 1976).

However, Understanding the presence and abundance of parasites in cultured fish species is crucial for developing effective management strategies. This knowledge aids in implementing targeted interventions, such as appropriate stocking densities, water quality management, and timely therapeutic measures, to mitigate the adverse effects of parasitic infestations.

Aims and Objectives:

- To identify the parasites up to their species level.
- To determine the prevalence of parasitic infestations among the selected fish species.

Methodology:**Sample collection:**

Samples will be collected from the freshwater cultured fish farm of Boalkhali upazila. After collecting the samples in specific numbers for research, the samples will be immediately brought to the laboratory, Department of Fisheries, University of Chittagong and preserved in a freezer for further analysis.

Measurement of length and weight:

The total length of all the samples will be taken and accounted for during the study period by millimeter scale. Besides, the weight of individual specimens will be taken with the help of a “pan balance.”

Extraction of parasites:

All the samples will be carefully observed to find out any ectoparasites on the surface of the fish species. Also, each sample digestive tract will be extracted and collected in different Petri dishes with 0.7% NaCl solution for extracting parasites present in stomach and intestine using stereoscopic binocular dissecting microscope.

Preservation of samples:

The stomach and intestine will immediately preserve with 70% alcohol for further identification of parasites. On the other hand, identified parasite will be stored or preserved separately according their range. Some will be preserved in 70% ethyl alcohol and rest in lactophenol for clear visibility of parasites under microscope.

Identification:

The identification and confirmation of the parasites will be done by comparing the data obtained from the research with material found in international articles, books, literatures and the internet, as well as their taxonomy and characteristics.

Expected outcomes:

The research is expected to provide valuable insights into the abundance of parasites in different cultured fishes. This analysis will also help in understanding the relationship between fish species and abundance of parasites in freshwater ecosystem.

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